Physical Chemistry (I)

Lecture 08

In the Last Class

- Thermochemistry
  - Standard enthalpy
  - Standard reaction enthalpy
  - Standard enthalpy of formation
  - Hess's law and Kirchhoff's law
- The second law of thermodynamics
  - Heat engine
  - Two versions of the second law

Thermochemistry

"The study of the energy transferred as heat during the course of chemical reactions"

$$\underbrace{\mathrm{A} + \mathrm{B}}_{\text{state } i} \longrightarrow \underbrace{\mathrm{C} + \mathrm{D}}_{\text{state } f}$$

$$q_V = \Delta U$$

$$q_p = \Delta H$$

Standard Enthalpy, $\boldsymbol{H^{\ominus}}$

Enthalpy H of a substance in its standard state

- Standard state of a substance: its pure form at 1 bar
  - Defined for a specific temperature
  - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^{\ominus} = H_f^{\ominus} - H_i^{\ominus}$

Review

Standard Reaction Enthalpy

$$\nu_\mathrm{A}\mathrm{A} + \nu_\mathrm{B}\mathrm{B} \longrightarrow \nu_\mathrm{C}\mathrm{C} + \nu_\mathrm{D}\mathrm{D}$$

- Standard reaction enthalpy $\Delta_\mathrm{r}H^\ominus$: standard enthalpy change $\Delta H^\ominus$ for the reaction per …
  - ν_A moles of consumed A
  - ν_B moles of consumed B
  - ν_C moles of produced C
  - ν_D moles of produced D

Hess's Law

"The standard reaction enthalpy is

the sum of the values for the individual reactions

into which the overall reaction may be divided."

This holds because H is a state function

Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_f H^{\ominus}$ of a substance: $\Delta_r H^{\ominus}$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

e.g. $\Delta_f H^{\ominus}$ of liquid benzene at 298 K: +49.0 kJ mol^{-1}

$$6C(s, \text{graphite}) + 3H_2(g) \longrightarrow C_6H_6(l)$$

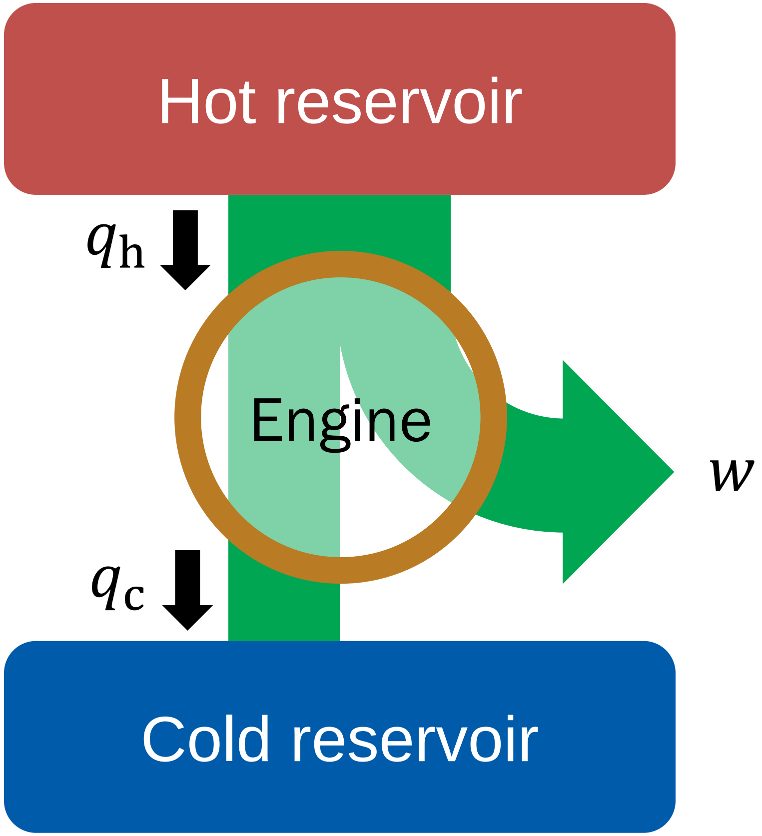
Kirchhoff's Law

$$\Delta_r H^{\ominus}(T_f) = \Delta_r H^{\ominus}(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^{\ominus}\, dT$$

where

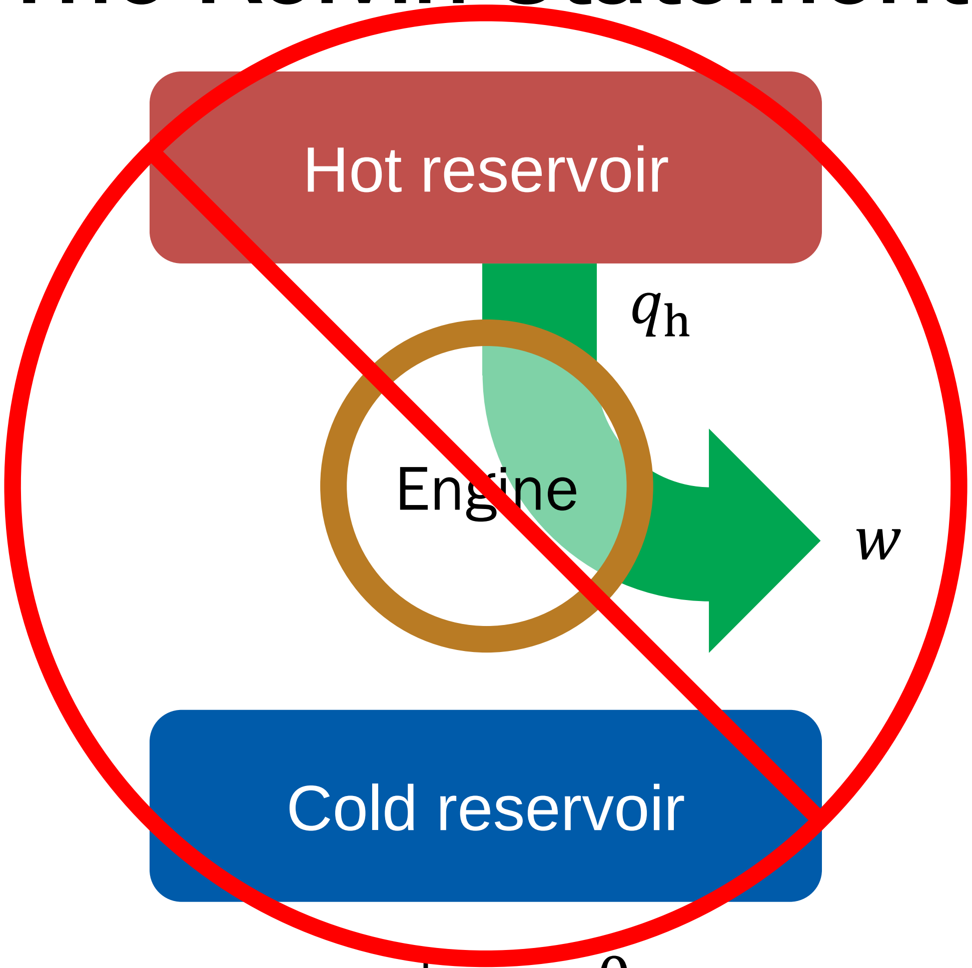
$$\Delta_r C_p^{\ominus} = \sum_{\text{products } i} \nu(i) C_{p,m}^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j) C_{p,m}^{\ominus}(j)$$

Heat Engine



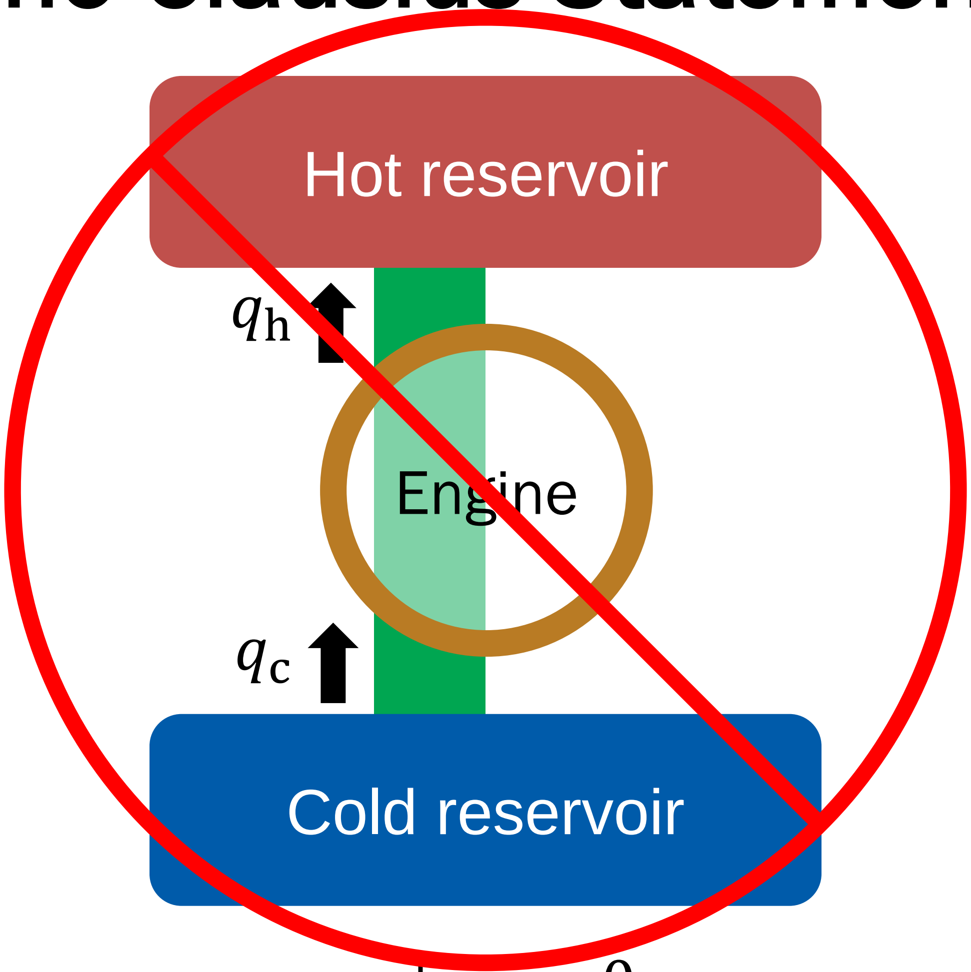
$$q_{\mathrm{h}} + q_{\mathrm{c}} + w = 0$$

The Kelvin Statement



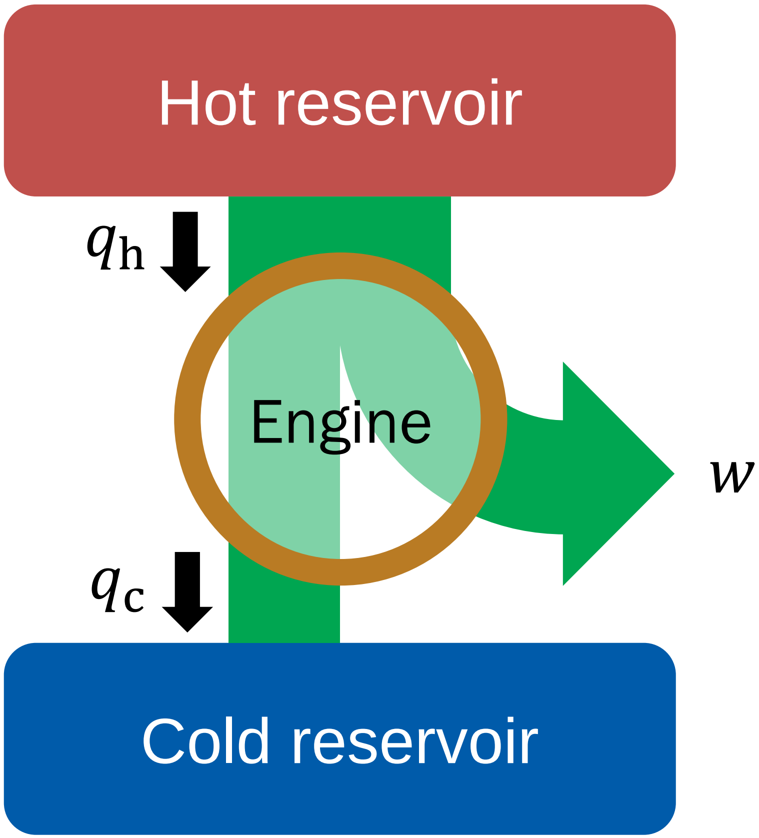
$$q_{\mathrm{h}} + w = 0$$

The Clausius Statement



$$q_h + q_c = 0$$

Efficiency of the Engine



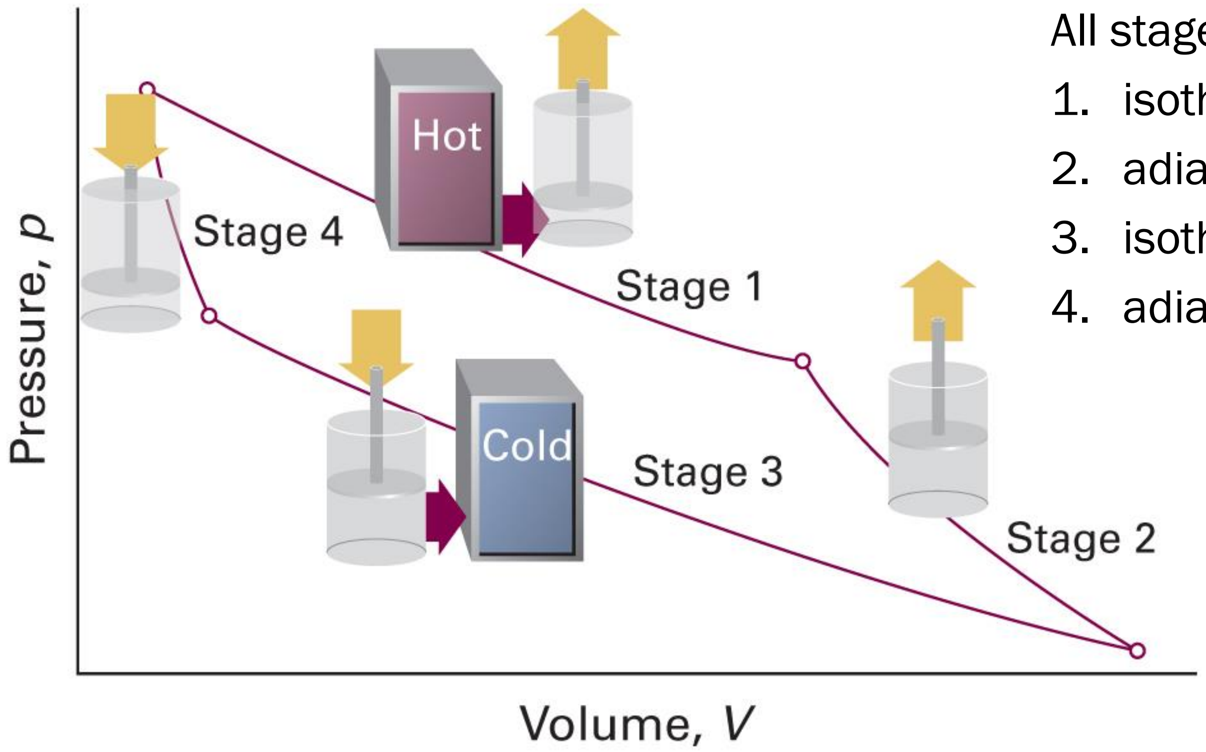
$$\eta = \frac{|w|}{|q_{\mathrm{h}}|}$$

Hence,

$$\eta = \frac{|q_{\mathrm{h}}| - |q_{\mathrm{c}}|}{|q_{\mathrm{h}}|} = 1 - \frac{|q_c|}{|q_h|}$$

$$q_{\mathrm{h}} + q_{\mathrm{c}} + w = 0$$

Carnot Cycle



All stages are reversible;

1. isothermal expansion
2. adiabatic expansion
3. isothermal compression
4. adiabatic compression

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.8)

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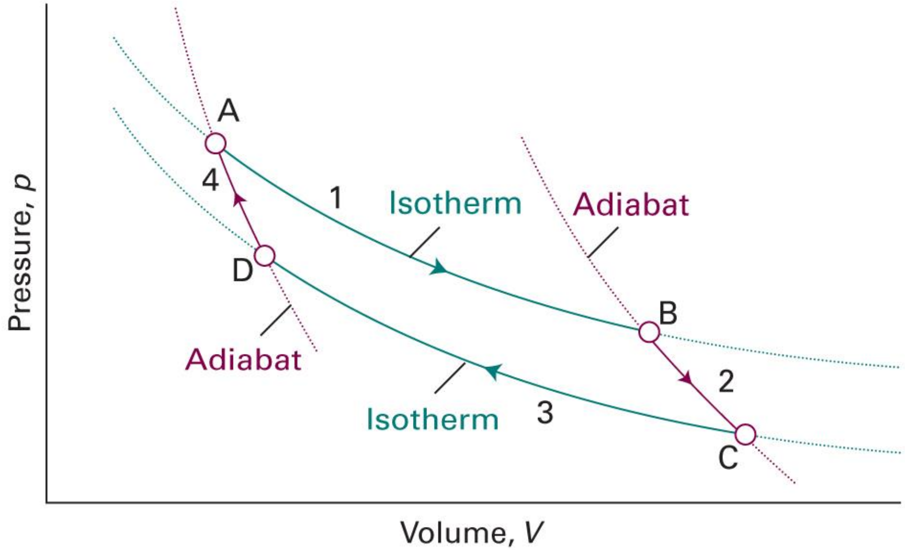
Perfect Gas in Carnot Engine



$T = T_\mathrm{h}$

$T = T_\mathrm{c}$

1. Isothermal expansion (at T_h)

$$w_1 = -nRT_\mathrm{h} \ln(V_\mathrm{B}/V_\mathrm{A})$$

Since $\Delta U = 0$, $q_1 = nRT_\mathrm{h} \ln(V_\mathrm{B}/V_\mathrm{A})$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.9)

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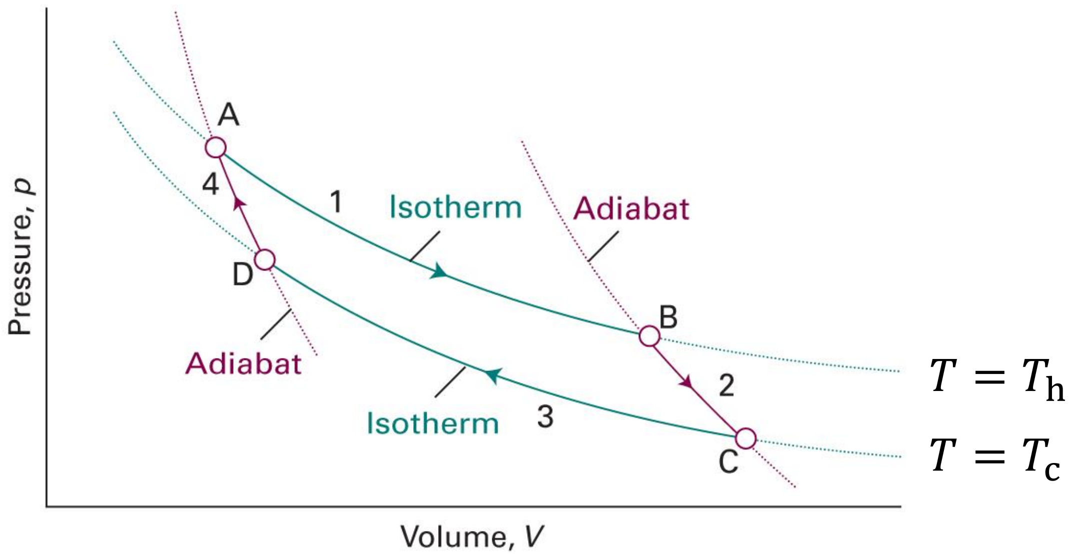
Perfect Gas in Carnot Engine



2. Adiabatic expansion

$$q_2 = 0$$

$$w_2 = C_V(T_c - T_h)$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.9)

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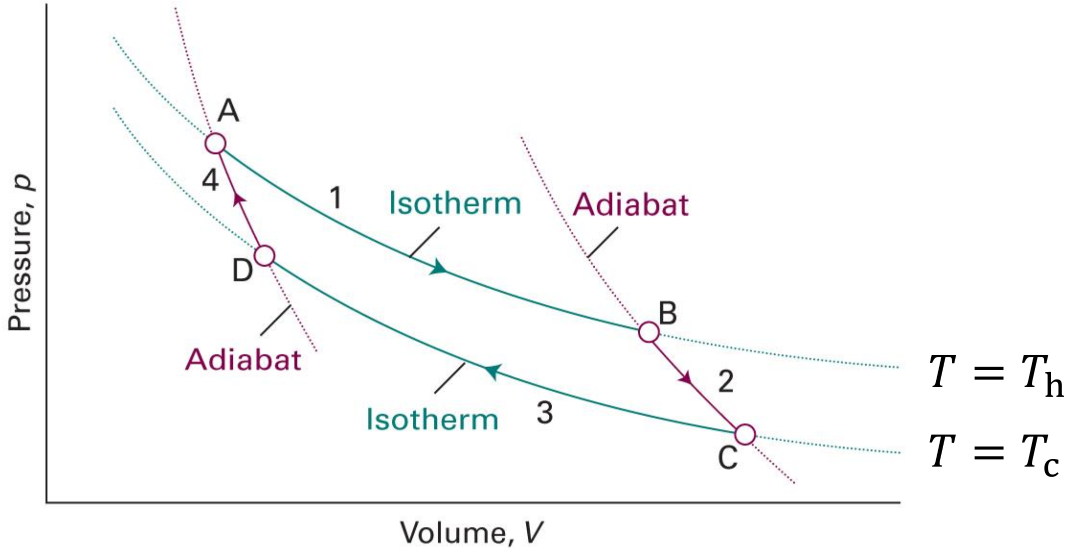
Perfect Gas in Carnot Engine



3. Isothermal compression (at T_c)

$$w_3 = -nRT_c \ln(V_D/V_C)$$

Since $\Delta U = 0$, $q_3 = nRT_c \ln(V_D/V_C)$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.9)

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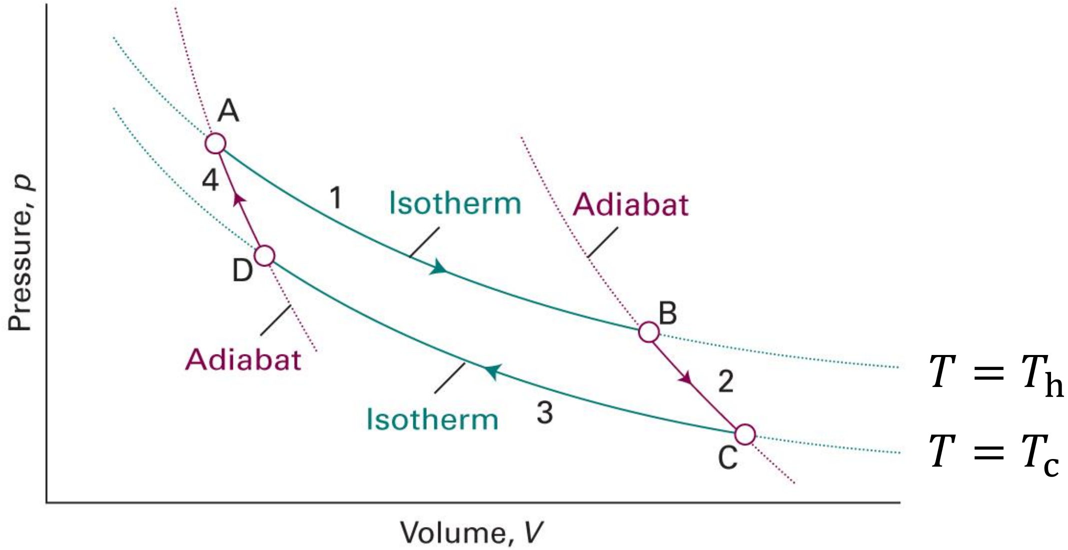
Perfect Gas in Carnot Engine



4. Adiabatic compression

$$q_4 = 0$$

$$w_4 = C_V(T_h - T_c)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.9)

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After One Cycle …

- Internal energy change $\Delta U = 0$ (state function)
- Enthalpy change $\Delta H = 0$ (state function)

- Heat

$$q = q_1 + q_2 + q_3 + q_4 = nRT_\mathrm{h}\ln(V_\mathrm{B}/V_\mathrm{A}) + nRT_\mathrm{c}\ln(V_\mathrm{D}/V_\mathrm{C})$$

- Work

$$w = w_1 + w_2 + w_3 + w_4$$

$$w = -nRT_\mathrm{h}\ln(V_\mathrm{B}/V_\mathrm{A}) - nRT_\mathrm{c}\ln(V_\mathrm{D}/V_\mathrm{C}) = -q$$

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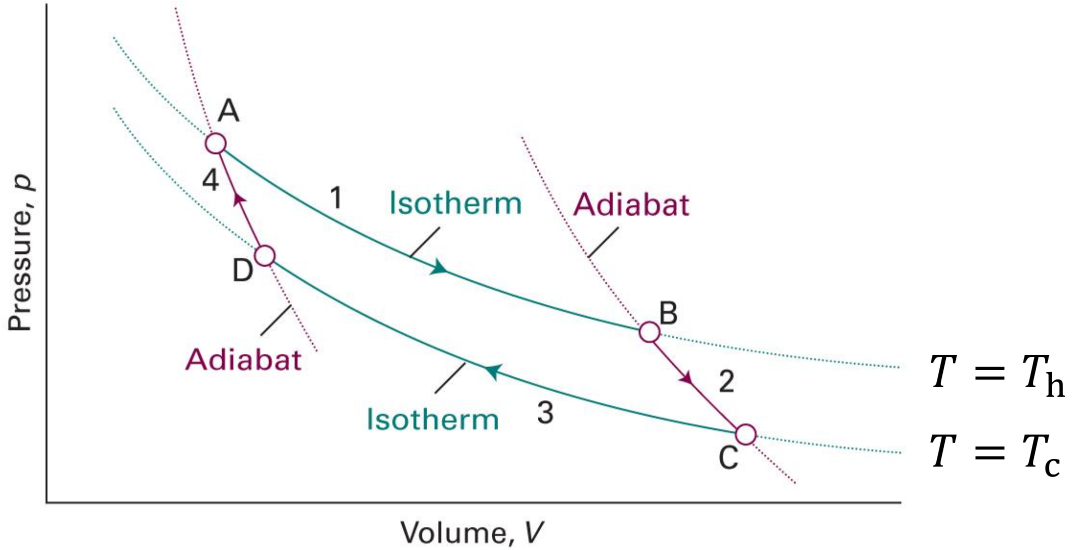
Use of Adiabats



2. Adiabatic expansion: $V_\mathrm{B} T_\mathrm{h}^c = V_\mathrm{C} T_\mathrm{c}^c$ $(c = C_V/nR)$

4. Adiabatic compression: $V_\mathrm{D} T_\mathrm{c}^c = V_\mathrm{A} T_\mathrm{h}^c$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11^{th} ed., Figure 3A.9)

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Heat and Work

Since $V_{\mathrm{B}}T_{\mathrm{h}}^{c} = V_{\mathrm{C}}T_{\mathrm{c}}^{c}$ and $V_{\mathrm{D}}T_{\mathrm{c}}^{c} = V_{\mathrm{A}}T_{\mathrm{h}}^{c}$,

$$\frac{V_{\mathrm{B}}}{V_{\mathrm{C}}} = \left(\frac{T_{\mathrm{c}}}{T_{\mathrm{h}}}\right)^{c} = \frac{V_{\mathrm{A}}}{V_{\mathrm{D}}}$$

Hence,

$$q = nRT_{\mathrm{h}} \ln\left(\frac{V_{\mathrm{B}}}{V_{\mathrm{A}}}\right) + nRT_{\mathrm{c}} \ln\left(\frac{V_{\mathrm{D}}}{V_{\mathrm{C}}}\right) = nR(T_h - T_c) \ln\left(\frac{V_{\mathrm{B}}}{V_{\mathrm{A}}}\right)$$

The work is

$$w = -q = -nR(T_h - T_c) \ln\left(\frac{V_{\mathrm{B}}}{V_{\mathrm{A}}}\right)$$

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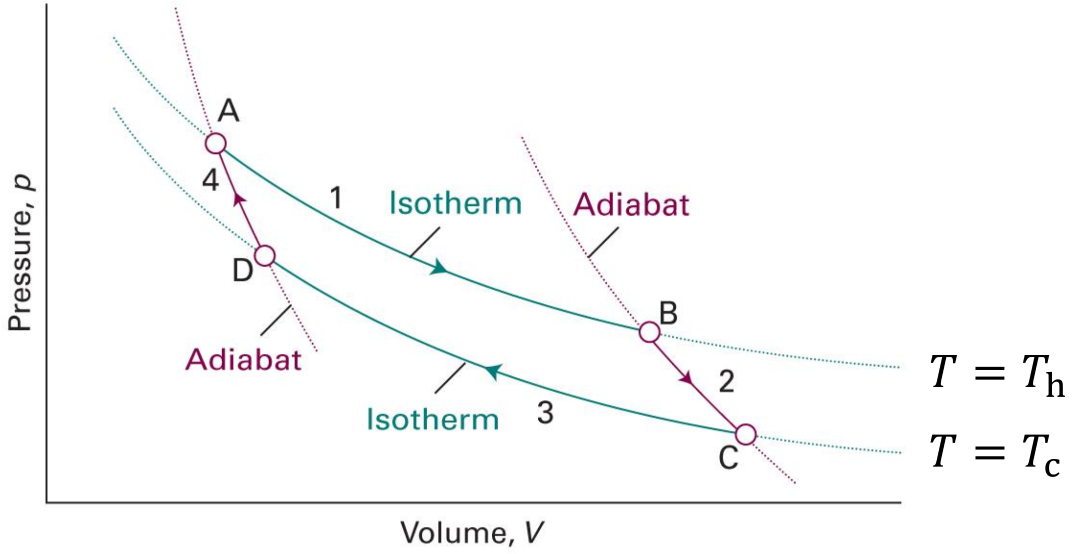
Efficiency of Carnot Engine



$$\eta = \frac{|w|}{|q_h|} = \frac{|w|}{|q_1|}$$

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Efficiency of Carnot Engine

$$\eta = \frac{|w|}{|q_\mathrm{h}|} = \frac{|w|}{|q_1|}$$

For $w = -nR(T_\mathrm{h} - T_\mathrm{c})\ln(V_\mathrm{B}/V_\mathrm{A})$ and $q_1 = nRT_\mathrm{h}\ln(V_\mathrm{B}/V_\mathrm{A})$,

$$\eta = \frac{nR(T_\mathrm{h} - T_\mathrm{c})\ln(V_\mathrm{B}/V_\mathrm{A})}{nRT_\mathrm{h}\ln(V_\mathrm{B}/V_\mathrm{A})} = \frac{T_\mathrm{h} - T_\mathrm{c}}{T_\mathrm{h}} = 1 - \frac{T_\mathrm{c}}{T_\mathrm{h}}$$

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Carnot's Theorem

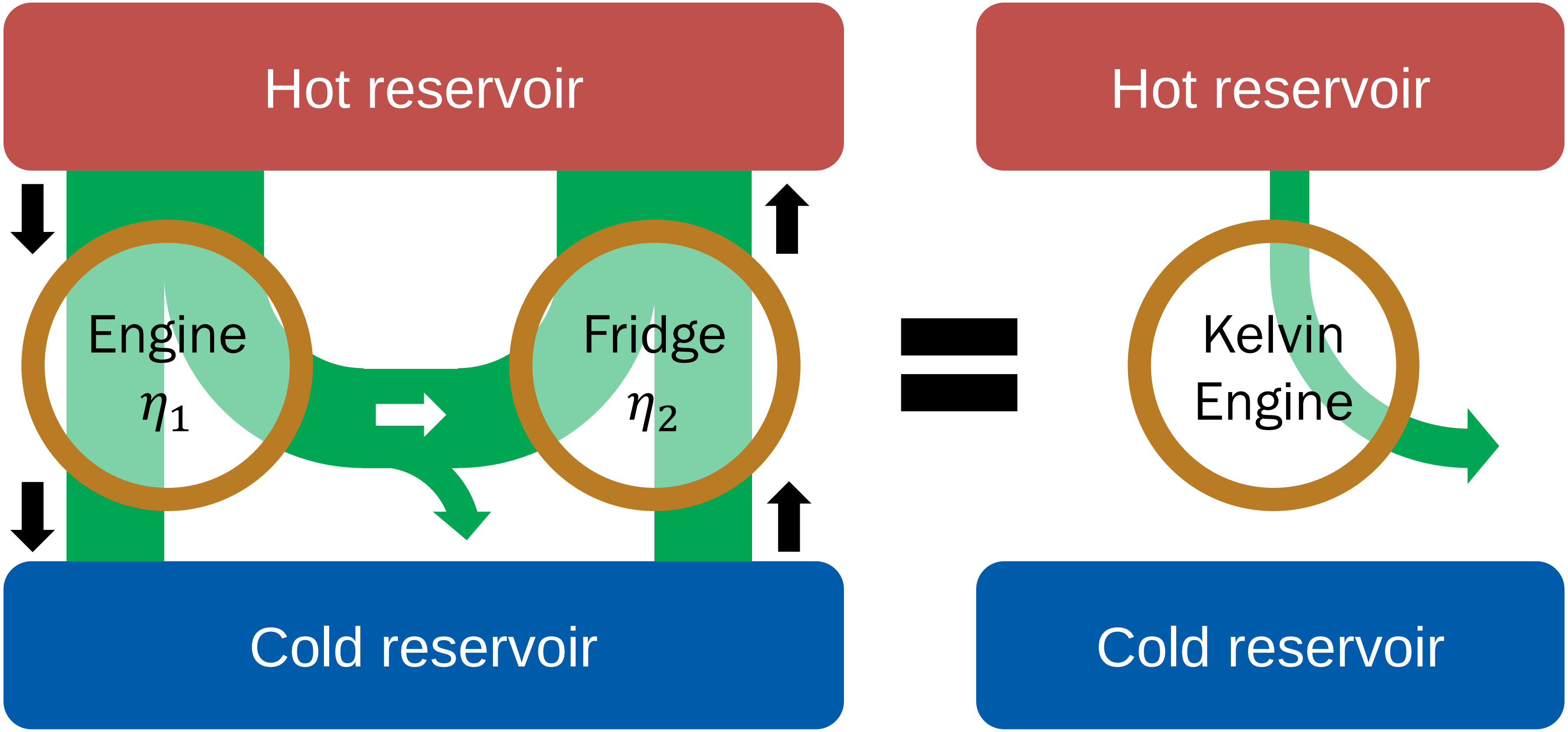
$\eta = 1 - T_\mathrm{c}/T_\mathrm{h}$ (perfect-gas reversible Carnot engine)

"The efficiency of a reversible Carnot engine depends only on the temperatures of the reservoirs."

$\longrightarrow$ independent of the working material

$\eta = 1 - T_\mathrm{c}/T_\mathrm{h}$ (any reversible Carnot engine)

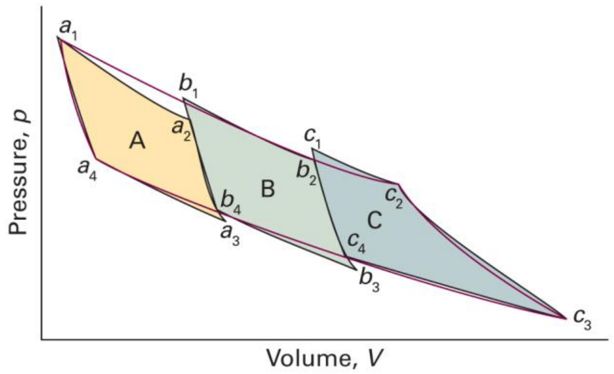
Proof of Equivalence



(if $\eta_1 > \eta_2$)

General Reversible Engine

"All reversible engines have the same efficiency regardless of their construction"



$\eta = 1 - T_\mathrm{c}/T_\mathrm{h}$ (any reversible engine)

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.12)

Brief Illustration 3A.3

A certain power station operates with superheated steam at 300 °C (T_{h} = 573 K) and discharges the waste heat into the environment at 20 °C (T_{c} = 293 K). What is its theoretical efficiency?

$$\eta = 1 - \frac{293\ \mathrm{K}}{573\ \mathrm{K}} = 0.489 = 48.9\%$$

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One More State Function

$$\eta = 1 - \frac{|q_\mathrm{c}|}{|q_\mathrm{h}|} = 1 + \frac{q_\mathrm{c}}{q_\mathrm{h}}$$

For a reversible engine,

$$\eta = 1 - \frac{T_\mathrm{c}}{T_\mathrm{h}}$$

Hence, for a reversible cycle,

$$\frac{q_\mathrm{h}}{T_\mathrm{h}} + \frac{q_\mathrm{c}}{T_\mathrm{c}} = \Delta\left(\frac{q}{T}\right) = 0$$

Entropy

For a reversible cycle,

$$\Delta\left(\frac{q_{\mathrm{rev}}}{T}\right) = 0$$

which implies that q_{rev}/T is a state function

Thermodynamic definition of entropy:

$$dS = \frac{\delta q_{\mathrm{rev}}}{T}$$

Entropy Calculation

$$dS = \frac{\delta q_{\mathrm{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\mathrm{state}\ i}^{\mathrm{state}\ f} \frac{\delta q_{\mathrm{rev}}}{T}$$

- Molar entropy change $\Delta S_{\mathrm{m}} = \Delta S/n$

Example 3A.1

Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f.

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

Since it is an isothermal change,

$$\Delta S = \frac{1}{T}\int_{\text{state } i}^{\text{state } f} \delta q_{\text{rev}} = \frac{q_{\text{rev}}}{T}$$

For an isothermal change of perfect gas,

$$\Delta U = q + w = 0 \text{ and } w = -nRT\ln\left(V_f/V_i\right)$$

Example 3A.1

Since $q = nRT\ln(V_f/V_i)$,

$$\Delta S = \frac{q_{\mathrm{rev}}}{T} = nR\ln\left(\frac{V_f}{V_i}\right)$$

Heat Transfer to Surroundings

Consider an infinitesimal transfer of heat

$$\delta q = -\delta q_{\mathrm{sur}}$$

The surroundings are extremely large, so

$$\delta q_{\mathrm{sur}} = dU_{\mathrm{sur}} \text{ (constant volume)}$$

$$\delta q_{\mathrm{sur}} = dH_{\mathrm{sur}} \text{ (constant pressure)}$$

Since $dH = dU + d(pV)$, this implies $d(pV)_{\mathrm{sur}} \approx 0$

Entropy of Surroundings

U and H are state functions, so δq_{sur} is independent of reversibility:

$$dS_{\mathrm{sur}} = \frac{\delta q_{\mathrm{sur,rev}}}{T_{\mathrm{sur}}} = \frac{\delta q_{\mathrm{sur}}}{T_{\mathrm{sur}}}$$

As the temperature is invariant,

$$\Delta S_{\mathrm{sur}} = \frac{q_{\mathrm{sur}}}{T_{\mathrm{sur}}}$$

Work for Irreversible Change

For an irreversible change,

$$|\delta w_{\mathrm{irrev}}| < |\delta w_{\mathrm{rev}}|$$

As $dw < 0$,

$$\delta w_{\mathrm{irrev}} > \delta w_{\mathrm{rev}}$$

In general,

$$\delta w \geq \delta w_{\mathrm{rev}}$$

Clausius Inequality

$$\delta w \geq \delta w_{\text{rev}}$$

From the first law of thermodynamics,

$$dU = \delta q + \delta w = \delta q_{\text{rev}} + \delta w_{\text{rev}}$$

which leads to

$$\delta q_{\text{rev}} - \delta q = \delta w - \delta w_{\text{rev}} \geq 0$$

Hence,

$$dS = \frac{\delta q_{\text{rev}}}{T} \geq \frac{\delta q}{T}$$

Isolated System

For an isolated system, $\delta q = 0$

$$dS \geq \frac{\delta q}{T} = 0$$

3rd version of the second law of thermodynamics:

"The entropy of an isolated system never decreases."

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Entropy and Spontaneity

For an isolated system,

$$dS \geq 0$$

- Reversible change: $dS = 0$
  - The change can be reversed
  - Achieved only at equilibrium
- Irreversible change: $dS > 0$
  - The change cannot be reversed
  - Arrow of time: spontaneous change

Fundamental Equation

From the first law,

$$dU = \delta q + \delta w$$

For a reversible change,

$$dU = \delta q_{\mathrm{rev}} + \delta w_{\mathrm{rev}} = TdS - pdV$$

In fact, for both reversible and irreversible changes,

$$dU = TdS - pdV$$

Natural Independent Variables

$$dU = TdS - pdV$$

$\rightarrow S$ and V are natural independent variables of U

$$U = U(S, V)$$

As U is a state function,

$$dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$$

Topic 3E

Thermodynamics Definitions

$$dU = TdS - pdV$$

As $U = U(S, V)$ is a state function,

$$dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_V$$

$$p = -\left(\frac{\partial U}{\partial V}\right)_S$$

Examples

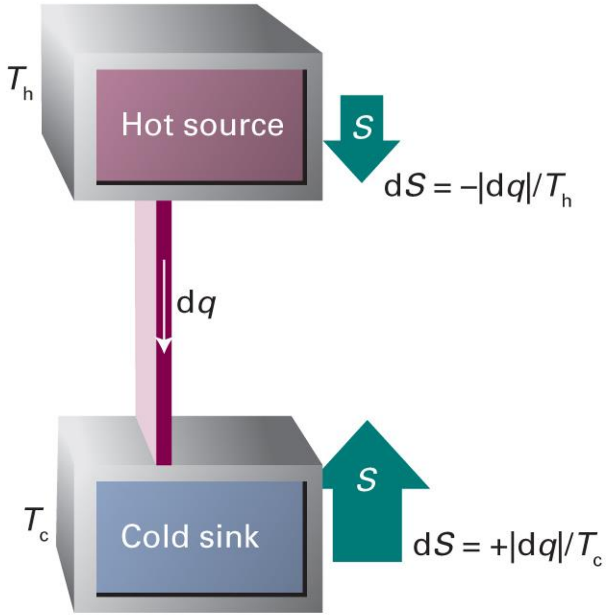
Heat transfer

Expansion (reversible vs. free)

Phase transitions

Heating

Heat Transfer



Transfer of heat $|\delta q|$: h → c

$$dS_{\mathrm{h}} = -|\delta q|/T_{\mathrm{h}}$$

$$dS_{\mathrm{c}} = +|\delta q|/T_{\mathrm{c}}$$

The total entropy change is

$$dS = dS_{\mathrm{h}} + dS_{\mathrm{c}} = |\delta q|\left(\frac{1}{T_{\mathrm{c}}} - \frac{1}{T_{\mathrm{h}}}\right)$$

- Equilibrium: $dS = 0$ or $T_{\mathrm{h}} = T_{\mathrm{c}}$
- Spontaneity: $dS > 0$ or $T_{\mathrm{h}} > T_{\mathrm{c}}$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.13)

Reversible Expansion

$$\delta q = -\delta q_{\mathrm{sur}}$$

$$\Delta S = \frac{q_{\mathrm{rev}}}{T}$$

$$\Delta S_{\mathrm{sur}} = \frac{q_{\mathrm{sur}}}{T} = -\frac{q_{\mathrm{rev}}}{T} = -\Delta S$$

Hence,

$$\Delta S_{\mathrm{total}} = \Delta S + \Delta S_{\mathrm{sur}} = 0$$

Free Expansion

For a perfect gas, isothermal expansion accompanies with

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right)$$

However, $w = 0$ (free expansion) and $\Delta U = 0$ (perfect gas)

$$q = q_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = 0$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} > 0$$

Reversible Phase Transition

Normal transition temperature T_{trs}: the temperature at which two phases are in equilibrium at 1 atm

If the system is at T_{trs} and 1 atm,

$$q = \Delta_{\mathrm{trs}}H \text{ (constant pressure)}$$

$$\Delta_{\mathrm{trs}}S = \Delta_{\mathrm{trs}}H/T_{\mathrm{trs}}$$

$$\Delta_{\mathrm{trs}}S_{\mathrm{sur}} = -\Delta_{\mathrm{trs}}H/T_{\mathrm{trs}}$$

$$\Delta_{\mathrm{trs}}S_{\mathrm{total}} = \Delta_{\mathrm{trs}}S + \Delta_{\mathrm{trs}}S_{\mathrm{sur}} = 0$$

Vaporization Data

	$\Delta_{vap}H^{\ominus}$ / (kJ mol^{-1})	$\Delta_{vap}S^{\ominus}$ / (J K^{-1} mol^{-1})	T_b / K
Benzene	30.8	87.2	353
CCl_4	30	85.8	350
Cyclohexane	30.1	85.1	354
H_2S	18.7	87.9	213
Methane	8.18	73.2	112
Water	40.7	109.1	373

Trouton's rule

Trouton's Rule

"A wide range of liquids give approximately the same standard entropy of vaporization (about 85 J K^{-1} mol^{-1})"

liquid $\longrightarrow$ gas: a huge change of volume

For a perfect gas, $\Delta S = nR\ln\left(V_f/V_i\right)$